

2. (15 puntos) En el circuito de condensadores de la Fig. 2, considere que $C_1 = 6 [pF]$, $C_5 = 4 [pF]$ y $C_2 = 8 [pF]$. C_3 es un condensador de placas planas de área $A = 18 [cm^2]$ y distancia entre ellas de $d = 3 [mm]$, relleno de PVC ($\kappa = 3$). C_4 es un condensador cilíndrico de largo $l = 5 [cm]$, radio interior $a = 3 [mm]$ y radio exterior $b = 5 [mm]$, relleno con mica ($\kappa = 4$).

- (a) Determine la capacitancia equivalente C_{eq} del circuito.
- (b) Si $V_{ab} = 24 [V]$ determine la carga almacenada en cada condensador.

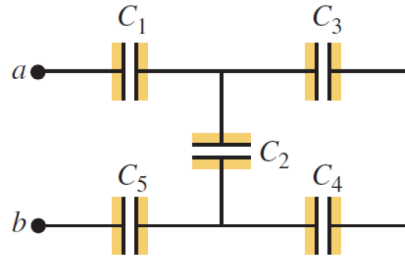
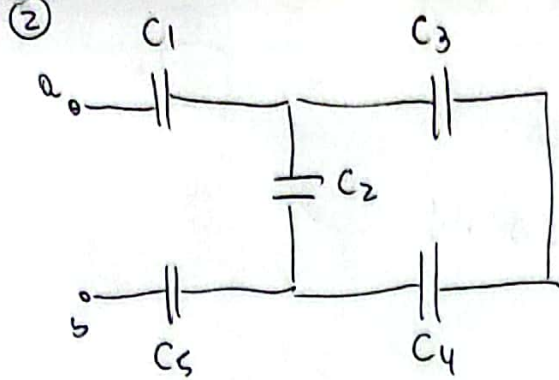


Figura 2



$$\frac{1}{C_a} = \frac{1}{C_3} + \frac{1}{C_4}$$

$$\frac{10^{-12}}{C_a} = \frac{1}{15,93} + \frac{1}{21,77}$$

$$C_a = 9,199 \text{ pF} \quad (1)$$

$$C_b = C_2 + C_a$$

$$= 8 \cdot 10^{-12} + 9,199 \cdot 10^{-12}$$

$$= 17,199 \text{ pF} \quad (1)$$

$$\frac{10^{-12}}{C_{eq}} = \frac{1}{6} + \frac{1}{4} + \frac{1}{17,199}$$

$$C_{eq} = 2,106 \text{ pF} \quad (3)$$

$$C_3 = K \epsilon_0 \frac{A}{d}$$

$$= 3 \cdot \epsilon_0 \cdot \frac{18 \cdot (0,01)^2}{3 \cdot 10^{-3}}$$

$$= 15,93 \text{ pF} \quad (1)$$

$$C_4 = K 2\pi \epsilon_0 \frac{l}{\ln(b/a)}$$

$$= 8\pi \epsilon_0 \frac{0,05}{\ln\left(\frac{5 \cdot 10^{-3}}{3 \cdot 10^{-3}}\right)}$$

$$= 21,77 \text{ pF} \quad (1)$$

$$C_{eq} = \frac{Q}{V_{ab}}$$

$$\boxed{q_1 = q_5 = q_3}$$

 EN SERIE

$$Q = C_{eq} \cdot V_{ab}$$

$$= 2,106 \cdot 10^{-12} \cdot 24$$

(3)

$$q_1 = 50,55 \text{ [pC]}$$

$$q_5 = 50,55 \text{ [pC]}$$

$$Q = 50,55 \text{ [pC]}$$

$$q_3 = 50,55 \text{ pC.}$$

$$V_2 = V_e = 2,939 \text{ [V]}$$

$$V_b = \frac{q_3}{C_B}$$

=>

$$= \frac{50,55 \cdot 10^{-12}}{17,199 \cdot 10^{-12}}$$

$$= 2,939 \text{ [V]}$$

$$q_2 = V_2 \cdot C_2$$

$$= 2,939 \cdot 8 \cdot 10^{-12}$$

(2)

$$= 23,51 \text{ [pC]}$$

$$V_e = \frac{q_A}{C_A}$$

$$q_A = V_e \cdot C_A$$

$$= 2,939 \cdot 9,199 \cdot 10^{-12}$$

$$= 27,03 \text{ [pC]}$$

=>

$$q_A = q_3 = q_4$$

$$\boxed{q_3 = q_4} \text{ en serie}$$

(3)

$$q_3 = 27,03 \text{ [pC]}$$

$$q_4 = 27,03 \text{ [pC]}$$